

MINA SALALAH, Oman

WMO: 413120

Lat: 16.934N

Lon: 54.008E

Elev: 25

StdP: 101.03

Time Zone: 4.00 (E04)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.8	19.8	-2.3	3.1	23.9	0.0	3.8	26.5	10.8	22.8	9.4	23.5	2.7	290	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	3.7	33.9	18.3	32.6	19.3	31.7	21.1	28.2	30.1	27.7	29.9	27.3	29.6	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	23.7	29.6	27.1	22.9	29.4	26.7	22.3	29.1	91.5	30.1	88.9	30.0	87.1	29.7	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	6.6	5.5	16.3	35.5	1.3	1.8	15.4	36.9	14.7	37.9	14.0	39.0	13.0	40.3
			8.5	28.1	1.7	0.8	7.3	28.6	6.3	29.1	5.3	29.5	4.1	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	23.7	24.2	25.9	28.0	29.6	28.7	25.5	24.4	26.0	26.7	26.8	25.4	(6)
(7)		DBStd	2.28	1.53	1.79	1.91	1.40	1.16	1.46	1.35	0.96	1.61	1.46	1.42	1.54	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5926	426	397	494	539	606	561	479	447	481	517	503	477	(10)
(11)		CDD18.3	2884	167	164	236	289	348	311	221	188	231	259	253	219	(11)
(12)		CDH23.3	26886	899	1083	2219	3388	4522	3723	1475	749	1844	2606	2622	1757	(12)
(13)		CDH26.7	7662	78	148	538	1187	2056	1388	135	10	294	713	782	332	(13)
(14)	Wind	WSAvg	2.3	3.0	2.6	2.2	2.1	2.3	2.4	1.8	1.6	2.1	1.9	2.4	3.3	(14)
(15)	Precipitation	PrecAvg	118	5	3	7	7	12	6	22	19	8	6	11	10	(15)
(16)		PrecMax	278	12	11	43	75	161	55	55	41	69	52	97	45	(16)
(17)		PrecMin	32	1	1	0	1	0	0	7	4	1	1	0	1	(17)
(18)		PrecStd	58	3	2	12	13	33	10	12	9	12	11	19	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.7	31.2	34.2	34.7	35.3	35.0	30.1	27.8	34.1	33.2	33.2	31.5	(19)
(20)			MCWB	15.9	16.4	17.4	17.9	19.1	26.0	26.8	23.6	18.6	17.9	17.4	16.7	(20)
(21)		2%	DB	28.6	29.7	32.2	32.8	33.5	32.6	28.9	26.7	31.2	31.6	31.9	30.2	(21)
(22)			MCWB	15.9	15.9	17.1	17.8	20.0	27.7	26.3	23.9	17.5	19.1	17.3	16.5	(22)
(23)		5%	DB	27.6	28.5	30.5	31.5	32.2	31.4	28.0	26.1	29.0	30.6	30.7	29.3	(23)
(24)			MCWB	16.9	16.1	17.2	20.0	23.4	27.5	25.9	23.9	20.8	20.4	18.3	16.9	(24)
(25)		10%	DB	26.7	27.3	29.2	30.4	31.4	30.6	27.3	25.6	28.1	29.7	29.7	28.4	(25)
(26)			MCWB	18.4	18.5	19.9	23.2	24.8	27.2	25.4	23.8	22.8	21.9	20.1	18.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	24.0	26.1	28.5	29.6	28.6	27.0	26.0	26.7	26.4	25.8	23.9	(27)
(28)			MCD B	26.0	26.4	27.3	29.5	30.2	30.6	28.5	27.0	27.5	29.2	28.9	27.5	(28)
(29)		2%	WB	21.7	23.0	25.3	27.1	28.6	28.1	26.5	25.2	25.8	25.8	24.9	23.1	(29)
(30)			MCD B	25.6	26.1	27.6	29.4	30.3	30.0	27.9	26.0	27.3	28.9	28.3	27.1	(30)
(31)		5%	WB	21.2	22.4	24.5	26.3	27.7	27.7	26.0	24.8	25.3	25.3	24.2	22.5	(31)
(32)			MCD B	25.4	25.8	27.5	29.2	30.4	29.6	27.3	25.6	26.8	28.3	28.1	26.7	(32)
(33)		10%	WB	20.7	21.7	23.7	25.8	27.3	27.5	25.5	24.5	25.0	24.9	23.7	22.0	(33)
(34)			MCD B	25.1	25.5	27.2	28.9	30.2	29.4	26.6	25.3	26.5	27.8	27.8	26.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	5.6	4.8	3.7	2.9	2.0	1.9	2.9	5.3	6.3	6.0	(35)
(36)			MCDBR	6.3	7.0	6.8	6.5	4.8	4.2	2.8	2.6	4.0	6.5	7.3	6.7	(36)
(37)			MCWBR	3.5	4.1	3.7	3.1	1.9	1.6	1.2	1.2	1.6	3.4	4.0	3.2	(37)
(38)		5% WB	MCD BR	5.3	4.9	4.6	3.9	3.2	2.5	2.4	2.2	3.0	4.6	5.5	5.2	(38)
(39)			MCWBR	3.3	3.3	3.3	2.5	1.7	1.3	1.1	1.2	1.7	3.6	4.1	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.414	0.436	0.485	0.513	0.602	0.778	0.824	0.756	0.607	0.452	0.432	0.415	(40)	
(41)		taud	2.220	2.144	2.008	1.974	1.785	1.491	1.451	1.559	1.796	2.162	2.202	2.218	(41)	
(42)		Ebn at Noon	868	867	834	809	730	606	580	626	724	841	842	853	(42)	
(43)		Edh at Noon	135	153	181	188	224	297	310	280	220	148	137	132	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.46	6.27	6.97	7.37	7.37	6.56	5.59	5.66	6.21	6.42	5.62	5.24	(44)	
(45)		RadStd	0.14	0.16	0.25	0.20	0.26	0.33	0.31	0.20	0.28	0.16	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.15	N/A	-1.52	N/A	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page